

María Fernández

Senior R&D and AI Engineering Manager · PhD Researcher in AI & Computer Science

Computer vision and AI for biology and medicine · Data & AI strategy and architecture

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PROFILE

R&D and engineering leader with nearly 20 years turning Data & AI research and solutions into production systems for regulated industries. I lead multidisciplinary teams of researchers and engineers from problem definition to deployment, owning technical direction, delivery and stakeholder alignment.

PhD candidate at Universitat de Barcelona and member of the AIBA research group, working on artificial intelligence and computer vision applied to biology and medicine, building on master's training in biomedical data science and applied AI.

CORE EXPERTISE

R&D and engineering leadership: research roadmaps, building and mentoring multidisciplinary teams, research-to-production transfer, agile delivery, executive and client stakeholder management

AI for biology and medicine: medical image analysis, computer-aided diagnosis, multimodal models, biomedical signal processing, clinical omics, food computing, health data ethics and regulation

Computer vision and deep learning: PyTorch, TensorFlow, Keras; CNNs, transformers, VAEs, GANs, diffusion models, reinforcement learning

Generative and agentic AI: generative models for language and computer vision, foundation models, AI agents and multi-agent systems, agentic retrieval (RAG), model evaluation frameworks

MLOps and platforms: model evaluation, A/B testing, gradual rollout, monitoring; Azure, AWS, Google Cloud; big data and streaming architectures

RESEARCH

PhD Candidate, AI Research

Universitat de Barcelona

I am a PhD candidate at the Faculty of Mathematics and Computer Science, Universitat de Barcelona, and a member of the AIBA research group. My research focuses on artificial intelligence and computer vision applied to biology and medicine, and I supervise students in their research projects.

SELECTED RESEARCH PROJECTS

AI-driven Collision Cross Section (CCS) prediction for metabolomics

MSc Biomedical Data Science research placement · Universitat Rovira i Virgili

- Addressed CCS prediction from molecular structure, a key open problem for molecular identification in ion mobility spectrometry for metabolomics and lipidomics.
- Built a reproducible baseline from molecular fingerprints, m/z and adduct type, then generated SMILES-based molecular embeddings with Mol2Vec and ChemBERTa as complementary representations.
- Benchmarked descriptor-based against embedding-based models and analysed performance by adduct type to identify limitations and directions for improvement.

NutritionBERT: a physically and biochemically constrained masked Transformer for nutrient profile prediction

- Developing a masked Transformer that predicts the nutrient profile of foods from their molecular composition, with physical and biochemical constraints built into the model.
- Designed a leakage-controlled benchmark so that reported performance reflects generalisation to unseen foods rather than memorisation of near-duplicates.

EXPERIENCE

Principal Architect, Data & AI · Intelygenz (VASS Group), Madrid 01/2023 – 08/2026

- Led applied AI design and engineering for the company's generative AI platform, setting technical direction for teams delivering pilots and accelerators across sectors.
- Drove applied R&D from prototype to production across foundation models, AI agents and multi-agent systems.
- Introduced a model evaluation framework measuring task success, reasoning quality and reliability, now used to gate every release, with human-feedback loops for continuous improvement.
- Established MLOps practices for AI systems: version control, A/B testing, gradual rollout and production observability.
- Technical lead in executive workshops and proposals for enterprise clients, supporting multi-million-euro bids.

Principal Architect, Data & AI · Sopra Steria, Madrid 05/2021 – 01/2023

- Led and mentored interdisciplinary AI squads of engineers and data scientists, promoting knowledge transfer across teams.
- Defined AI adoption roadmaps and designed AI and cloud-native architectures for European finance and public-sector accounts.

Service Manager, Cloud · Santander Global Tech, Madrid 04/2020 – 05/2021

- Led cloud migration strategy across Google Cloud, AWS and Azure for internal banking clients, owning SLAs and service improvement.

Enterprise Architect, Data & AI · KPMG Lighthouse, Madrid 10/2017 – 04/2020

- Designed AI solutions and reference architectures reused across the Lighthouse global network.

Big Data Architect (contract) · Qvantel, Rome, and Cargill, Amsterdam 2017 – 2018

- Designed and delivered a production streaming data platform for Wind Tre, and the big data ingestion architecture for Cargill.

Implementation Project Manager · Santander UK, Milton Keynes 01/2015 – 06/2017

- Led planning and execution across development, architecture and analyst teams for the UK Banking Reform programme, managing risks and cross-team dependencies through to production roll-out.

Earlier career 2005 – 2015

- ISBAN, Santander Group (2008 – 2015): development team leader, then solution architect for data, business intelligence, fraud and risk; managed teams, estimates and delivery.
- Carrefour, Axpe Consulting and Glokal Consulting (2005 – 2008): software engineer, Java and C.

EDUCATION

PhD candidate, Mathematics and Computer Science Universitat de Barcelona · 2026 – present
Artificial intelligence and computer vision applied to biology and medicine; AIBA research group.

MSc, Biomedical Data Science Universitat Rovira i Virgili, School of Engineering
Data science for biomedical engineering and health: medical imaging, computer-aided diagnosis, neuroimaging, biomedical sensors and signal processing, clinical omics, computational epidemiology.

MSc, Advanced and Applied Artificial Intelligence Universitat de València
Deep learning, transformers, generative and diffusion models, reinforcement learning, multimodal models.

BSc (Hons) Computing, First Class Edinburgh Napier University

Deep Learning Expert Program MBIT School, Madrid

CERTIFICATIONS

Microsoft Certified: Azure AI Engineer Associate
SAFe Product Manager / Product Owner

LANGUAGES

Spanish (native)
English (C1)